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Management of water, sanitation and hygiene facilities in basic schools within the Mfantseman Municipality, Ghana

Kingsford Kobina Annan^{1*}, Simon Mariwah¹, Alice Annan² and Innocent Chirisa^{3,4,5}

*Correspondence:

Kingsford Kobina Annan
annankingsfordkobina@gmail.com

¹Department of Geography and
Regional Planning, University of
Cape Coast, Cape Coast, Central
Region, Ghana

²Department of Basic Education,
University of Cape Coast, Cape
Coast, Central Region, Ghana

³Department of Urban and
Regional Planning, University of
Johannesburg, Johannesburg,
South Africa

⁴Office of the Vice Chancellor,
Zimbabwe Ezekiel Guti University,
Bindura, Zimbabwe

⁵Department of Urban and
Regional Planning, University of
the Free State, Bloemfontein, South
Africa

Abstract

This study examined the management of Water, Sanitation, and Hygiene (WASH) facilities in public basic schools within the Mfantseman Municipality of Ghana. Using a convergent mixed-method design, data were collected from 368 students, 16 head teachers, and 4 institutional stakeholders. Results showed that 95.1% of rural and 87.5% of urban schools had toilet facilities, while 50% of rural and 75% of urban schools had access to water. All schools had handwashing facilities, though many were non-functional due to poor water supply. The Ghana Education Service (GES) and District Assemblies were the main providers of WASH infrastructure, while Parent Associations and communities supported with cleaning supplies and minor repairs. However, limited funding and coordination shifted maintenance responsibilities to teachers and pupils, 98% of whom handled cleaning. Only one school had a menstrual hygiene facility, with 89–94% of pupils reporting no sanitary bins. The study recommends that GES and District Assemblies increase financial commitment and supervision, the Ministry of Sanitation enforce hygiene standards, and Parent Associations promote gender-sensitive WASH planning. Clearly defined actor roles, consistent resource allocation, and community participation are essential for sustainable WASH management and equitable learning environments in Ghanaian schools.

Keywords Management, Pupils, Water, Sanitation, And hygiene (WASH) facilities

1 Introduction

Despite United Nations (UN) efforts, millions worldwide face a grim reality with thirst and the indignity of inadequate sanitation. Globally, 785 million lack clean water and 2.3 billion lack proper sanitation [29]. Even in schools, 44% globally lack basic toilets and 56% lack handwashing facilities [27]. Africa faces a water and sanitation crisis in its schools; more than half its schools lack basic sanitation and handwashing facilities [27].

In Ghana, over 1.3 million pupils remain without adequate WASH facilities, with public basic schools being the most disadvantaged due to limited state investment [6, 12]. Unlike private schools that are often sustained by fees and private investment, public



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basic schools in Mfantseman depend almost entirely on government and community support, which leaves them vulnerable to recurring shortages of infrastructure and logistical resources. The Mfantseman Municipality is especially notable for its low sanitation coverage and poor academic performance, according to the District League Table by [26]. This dual challenge of weak sanitation infrastructure and underperformance in education makes Mfantseman a critical area for research. UNICEF and WHO [28] highlight that in schools where WASH facilities are lacking or mismanaged, pupils particularly girls are more likely to miss school or perform poorly [4, 6, 9]. The need for this research in the Mfantseman Municipality arises from the fact that the area has consistently recorded some of the lowest sanitation coverage rates in Ghana, which directly undermines the health and well-being of pupils.

Numerous studies have demonstrated that students benefit significantly from access to well-managed Water, Sanitation, and Hygiene (WASH) facilities. For instance, [11] reported that schools with properly maintained WASH infrastructure recorded higher levels of student attendance and improved academic performance. Similarly, studies conducted in Ghana and across sub-Saharan Africa have confirmed that poor WASH conditions are directly linked to increased absenteeism, particularly among girls, and a higher prevalence of sanitation-related diseases [1–3]. These findings collectively reinforce the argument that effective WASH management is essential for promoting health, equity, and educational achievement among schoolchildren.

At the national level, Ghana's National Sanitation Policy 2015 as referenced by [16], and the [14] were introduced to address these challenges by ensuring adequate sanitation and hygiene in schools. However, despite these policy commitments, implementation remains a major concern. For instance, a 2017 EMIS assessment found that more than 74,000 schools lacked washrooms [12], compelling over two million pupils to practice open defecation [1]. Open defecation exposes pupils to infectious diseases and severely compromises their safety, privacy, and dignity, particularly among adolescent girls during menstruation [3]. Even schools with existing WASH facilities frequently struggle with limited water supply, inadequate hygiene materials, and poor maintenance, rendering facilities unsafe or unusable [4]. These challenges hinder Ghana's progress toward achieving Sustainable Development Goals (SDGs) 4 and 6, which emphasize quality education and universal access to clean water and sanitation.

The root causes of poor WASH management in Ghanaian schools have been widely associated with financial constraints, insufficient hygiene education, weak community participation, and gender disparities [1, 3, 23]. Despite numerous interventions, the persistence of these structural and behavioural barriers underscores a critical knowledge and policy gap, particularly regarding how WASH facilities are managed at the local level, where resource scarcity and institutional challenges are most pronounced.

It is against this backdrop that the Mfantseman Municipality was selected as the study area. The municipality presents a microcosm of Ghana's broader sanitation and education challenges. According to [26], Mfantseman consistently ranks among the districts with the lowest sanitation coverage and poorest academic performance. The area is characterized by a combination of coastal and rural settlements that depend heavily on government and community initiatives for school infrastructure provision, yet face chronic funding limitations and weak institutional oversight. The municipality's

sanitation conditions have direct implications for student health, attendance, and learning outcomes [7, 19].

Therefore, focusing on Mfantseman provides a unique opportunity to explore how WASH facilities are operated, maintained, and managed within the broader context of resource constraints and institutional challenges. Hence, this study aims to identify how schools currently manage their WASH facilities, including cleaning routines, maintenance, and supervision structures, explore the barriers to effective management, and propose evidence-based and policy-relevant strategies for improving facility maintenance and ensuring reliable access to safe water, sanitation, and hygiene services.

1.1 Sanitation and academic achievement by districts in Ghana

The pictures below show graphical representation of sanitation coverage and academic performance by district in Ghana. The red colours show district with poor sanitation coverage and low academic achievement while the green colours show district with good sanitation and better academic achievement, respectively. The pictures depict that Mfantseman Municipality is indicated with red colours for both sanitation and academic achievement (Fig. 1).

1.2 Theoretical framework

This research was guided by the Sanitation Behaviour Change Framework (SaniFOAM), which disaggregates sanitation-related behaviours into four interrelated elements: Focus, Opportunity, Ability, and Motivation [10]. Applying this model to WASH management in basic schools within the Mfantseman Municipality provides a structured lens through which both barriers and enablers of behaviour can be understood.

The focus of the framework specifies the target behaviour under study. In this research, the primary focus is on pupils and teachers in basic schools, specifically their interaction with the available WASH facilities, how toilets, water points, and handwashing stations are used, maintained, and managed. This clarity allows the study to examine not just infrastructure provision, but also the everyday practices that determine whether facilities function as intended.

Opportunity relates to the external conditions that enable or constrain the desired behaviour. In this context, it encompasses the availability of functional facilities, the

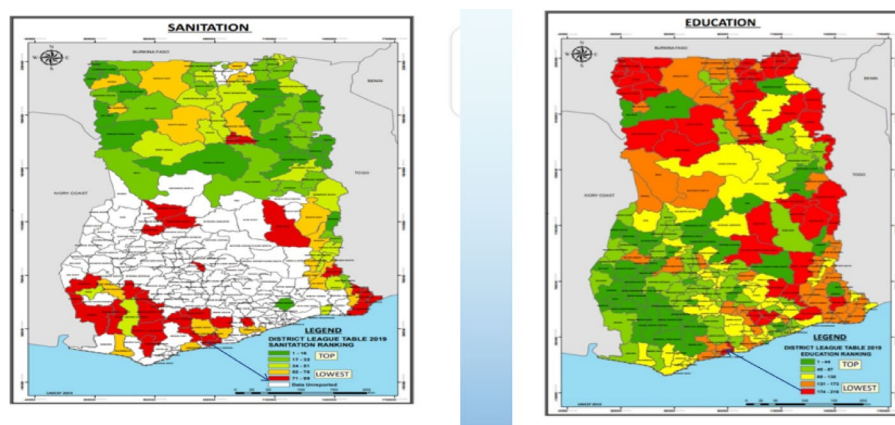


Fig. 1 Sanitation and academic achievement by districts in Ghana.

Source: UNICEF and CDD-Ghana [26]

presence of cleaning materials, and institutional support from school authorities or the Ghana Education Service. For example, schools with toilets but no reliable water supply face a structural barrier that undermines proper sanitation and hygiene behaviours. Similarly, schools that depend on students carrying water from home highlight how external limitations shape opportunity for WASH-related behaviours.

Ability captures the knowledge, skills, and resources individuals require to perform the behaviour. For pupils, this includes understanding why handwashing is important, knowing how to use Veronica buckets properly, and having access to soap and water at all times. For teachers and school administrators, ability is reflected in their capacity to manage WASH resources, enforce hygiene routines, and respond to breakdowns in facilities. If pupils lack awareness of good hygiene practices, or if teachers lack training in facility management, then behaviour change is constrained even when infrastructure is present.

Motivation refers to the internal and social drivers that influence whether or not the desired behaviour is adopted. In schools, motivation can be shaped by personal factors (such as a pupil's desire to avoid embarrassment or illness) and social influences (such as peer expectations, teacher supervision, or school-wide hygiene campaigns). The study findings, for example, the fact that all schools had handwashing facilities but not all had water, highlight that without strong motivational drivers (like awareness campaigns, incentives, or cultural reinforcement), facilities alone may not lead to consistent behavior (Fig. 2).

1.3 Study area and methodology

The Mfantseman Municipality, in Ghana's Central Region [15], is where this study was carried out. The Municipality is bordered to the northeast by Adjumako-Eyan-Essiam District, to the west by Abura-Asebu-Kwamankese District, to the east by Ekumfi District, and to the south by the Gulf of Guinea (Fig. 3).

The mixed-method approach that collects data using both quantitative and qualitative methods, was used in this study. For this research, a descriptive design in conjunction with a convergent mixed-method technique was employed. Combination of quantitative

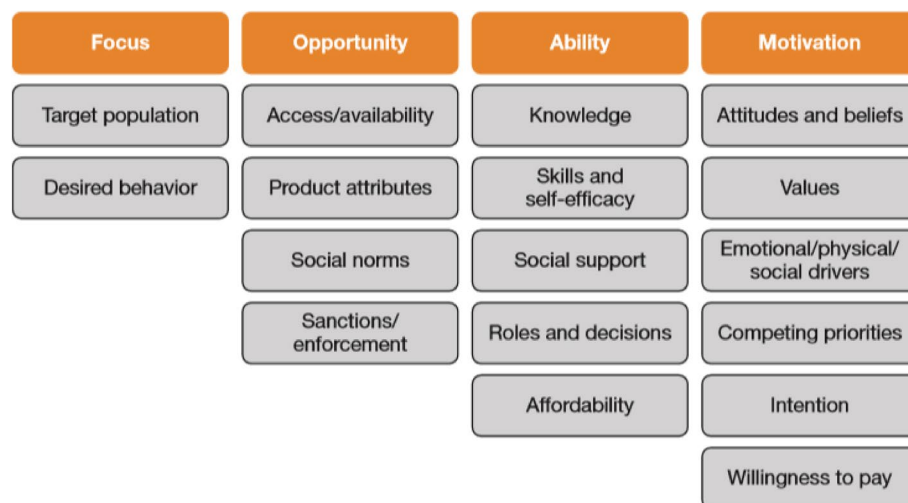


Fig. 2 Sanitation behaviour change framework (adopted).

Source: Devine [10]

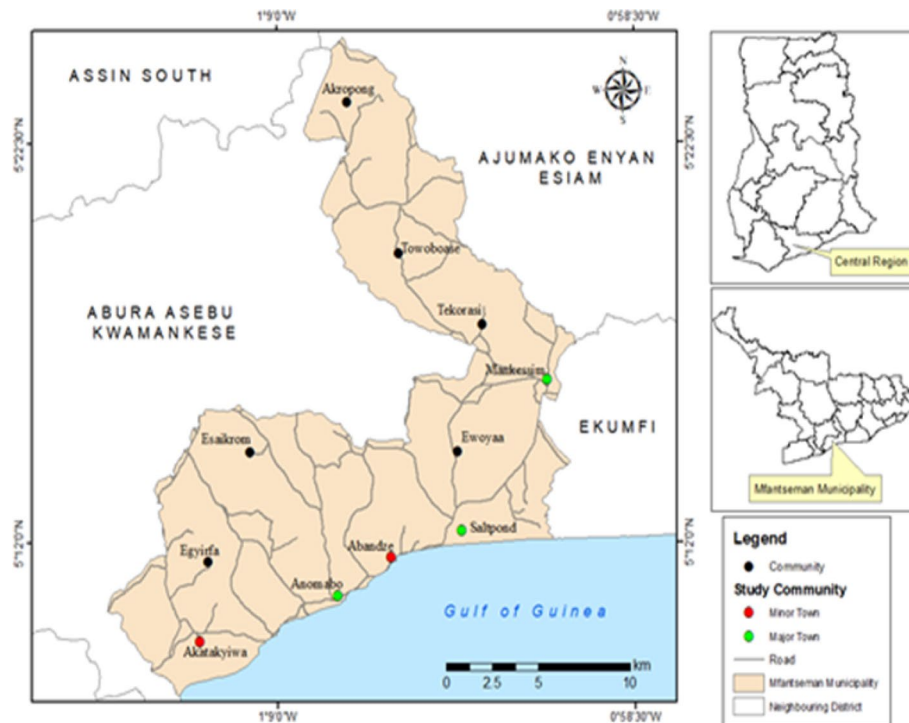


Fig. 3 Map of Mfantseman Municipality showing some minor and major vicinity.

Source: Department of Geography and Regional Planning, 2021

and qualitative data gave the researchers a comprehensive grasp of the research subject from different perspectives, and made it possible to cater for each other's flaws [8]. The convergent design allowed the researchers to collect quantitative and qualitative data concurrently, analyze them separately, and then merge the results to draw holistic conclusions [8]. Conversely, the descriptive approach is suitable for identifying, characterising, and elucidating people's behaviours and actions [17].

During the study period (February, 2021), the study's population comprised 8424 students in 86 public basic schools in the municipality. Using Hotjar's sample size calculator, a sample of 368 students was determined at a 95% confidence level and a 5% margin of error. A multi-stage sampling technique was used. First, the municipality was stratified into rural and urban zones. Eight rural and eight urban communities were randomly selected. From each community, one public basic school was chosen by simple random sampling. This produced a total of 16 schools.

To determine the number of students to be selected from each school, the total sample size (368) was divided equally among the 16 selected schools. This yielded 23 students per school ($368 \div 16 = 23$). This approach ensured proportional representation across all participating schools.

In each selected school, a class register was used as a sampling frame, and simple random sampling was applied to select pupils using the lottery method.

The inclusion criteria involved; pupils who had been enrolled in the school for at least one academic year, were of sound mind, and could provide meaningful responses. Pupils were selected across different classes and both male and female students were included to ensure gender inclusion and avoid bias.

The exclusion criteria were; pupils who had just transferred to the school (less than one year) or had cognitive impairments that could affect their ability to provide informed responses were excluded. This was to ensure data accuracy and consistency while upholding ethical standards. The exclusion was not meant to discriminate but to safeguard the integrity of the data collection process. The sampling approach also considered cultural, social, environmental, and economic contexts to capture a broad range of experiences with WASH services.

In addition to student respondents, 16 head teachers and four key institutional stakeholders (from GES, Environmental Health Unit, Public Works Department, and Health Directorate) were purposively selected for interviews to provide managerial and policy-level perspectives.

Data collection involved student questionnaires, key informant interviews, and structured observation checklists. The questionnaires captured quantitative data on facility use and satisfaction, while interviews provided qualitative insights into institutional responsibilities and policy implementation. Observations documented the physical state and usage of facilities.

Quantitative data were analysed using SPSS version 22, employing descriptive and inferential statistics, including chi-square tests. Qualitative data were transcribed and coded thematically, with photographic evidence providing contextual reinforcement. Ethical clearance was granted by the University of Cape Coast Institutional Review Board (UCCIRB/CHLS/2021/78), and informed consent was obtained from participants and guardians.

2 Results

Objective 1: Identify how schools currently manage their WASH facilities, including cleaning routines, maintenance, and supervision structures.

2.1 Available WASH facilities in the selected basic schools

Although the study showed that a majority of schools had WASH facilities, these were insufficient relative to student enrolment and international norms, such as the guideline of one toilet unit per 50 users recommended by global school sanitation frameworks [28]; and the Ghana National Sanitation Policy, 2015 as referenced by [16]. Within the Municipality, most public basic schools had access to toilets: 95.1% of rural schools and 87.5% of urban schools reported having toilet facilities. In terms of water access, 50% of pupils in rural schools and 75% in urban schools confirmed that their schools had a water supply. All rural schools (100%) surveyed had handwashing facilities, and nearly all urban schools (99.5%) did likewise. The data revealed that standpipes were the most commonly used water facility, Kumasi Ventilated Improved Pit (KVIP) toilets were the dominant sanitation option, and Veronica buckets were the primary handwashing devices in public basic schools in the Mfantseman Municipality (Table 1).

Further probing of the data revealed that schools without toilets often resorted to unsafe alternatives, such as open defecation along nearby beaches and in surrounding bushes. Such practices expose pupils and communities to significant health hazards, including diarrhea, cholera, and parasitic infections, while also compromising safety, privacy, and dignity, particularly for girls during menstruation (Fig. 4 and Table 2).

Table 1 Availability and functionality toilet facilities

Variables	Yes	No	Total	ASYMP.SIG		
Availability of toilet facility						
Rural	175 (95.1%)	9 (4.9%)	184 (100%)	0.010		
Urban	161(87.5%)	23(12.5%)	184 (100%)			
	Fully functional	Partially functional	Not functional	Total		
Functionality of toilet facility						
Rural	117 (66.9%)	39 (22.3%)	19 (10.9%)	175 (100%)	0.000	
Urban	129 (80%)	28 (17%)	4 (2.5%)	161 (100%)		
	KVIP	W/C	Total			
Type of toilet facility						
Rural	140 (80%)	35 (20%)	175 (100%)	0.000		
Urban	90 (55.9%)	71 (44.1%)	161 (100%)			
	Bush	Beach	Public toilet	Total		
Where pupils defecate in schools without a toilet						
Rural	1(11.1%)	6 (66.7%)	2 (22.2%)	9 (100%)	0.241	
Urban	8(34.8%)	11 (47.8%)	4 (17.4%)	23 (100%)		

Source: Field Survey (2021)

**Fig. 4** Toilet facilities mostly used in the basic schools**Table 2** Availability of functionality of water facilities

Table 2: Availability of functionality of water facilities						
Variables	Yes		No	Total	ASYMP.SIG	
Availability of water facility						
Rural	92 (50%)		92 (50%)	184 (100%)	0.000	
Urban	138 (75%)		46 (25%)	184 (100%)		
	Pipe	Well	Bole hole	Polytank	Total	
Type of water facility available in schools						
Rural	68 (73.9%)	13 (14.1%)	1(1.1%)	10 (10.9)	92 (100%)	0.000
Urban	115 (83.3%)	0 (0)	0 (0)	23 (16.7%)	138 (100%)	
	Fully functional	Partially functional	Not functional	Total		
Functionality of the water facility						
Rural	60 (65%)	9 (10%)	23 (25%)	92 (100%)	0.000	
Urban	97 (70.3%)	31(22.5%)	10 (7.2%)	138 (100%)		
	Open stream	Public pipe	Private ind	From home	Total	
Where students fetch water in schools without a water facility						
Rural	6 (6.5%)	56 (60.9%)	13 (14%)	17 (18.5%)	92 (100%)	0.000
Urban	0 (0)	30 (65%)	6 (13%)	10 (22%)	46 (100%)	

Source: Field Survey (2021)



Fig. 5 Water facilities used in the basic schools



Fig. 6 Handwashing Facilities mostly used in the basic schools

Variables	Yes	No	Total	ASYMP.SIG
Availability of hand washing facility in school				
Rural	184 (100%)	0 (0)	184 (100%)	0.317
Urban	183 (99.5%)	1 (0.5%)	184 (100%)	
Veronica bucket			Total	
Type of hand washing facility used in schools				
Rural	184 (100%)		184 (100%)	
Urban	183 (100%)		183 (100%)	

Source: Field survey (2021)

Additional investigation showed that in schools without water facilities, pupils and teachers relied on purchasing water from vendors, fetching from nearby households, or requiring students to carry water from home to school. These coping strategies are unsustainable and place an added financial and physical burden on families, while also limiting the consistent use of sanitation and hygiene infrastructure (Figs. 5, 6 and Table 3).

2.2 Distribution of toilet, water, and handwashing facilities across 16 basic schools in Mfantseman Municipality

In line with the [28] standards, this study applied international benchmarks to assess the availability of WASH facilities in schools. Schools were classified as having basic drinking water service if an improved water source was located on school premises.

For sanitation, schools were classified as having basic sanitation services only if usable, improved, and gender-segregated toilet facilities were available to pupils. Similarly, handwashing was considered adequate if facilities with both water and soap were available within the school compound. Schools failing to meet these minimum conditions were categorized as not having access to the respective WASH service.

The results presented in Table 4 reveal important disparities in the availability of WASH facilities across the 16 basic schools surveyed in the Mfantseman Municipality.

A majority of the schools (13 out of 16; 81.3%) had some form of toilet facility, while three schools (18.7%) lacked such infrastructure. Access to water was more uneven. Only nine schools (56.3%) had a reliable source of water within the premises, while seven schools (43.7%) reported having no water supply.

In contrast, handwashing facilities were reported in all 16 schools (100%). While this suggests progress in hygiene promotion, likely influenced by COVID-19-related interventions [30], these facilities are not always functional due to the absence of reliable water supply. This means that in some cases, the presence of toilets and handwashing infrastructure is symbolic rather than practical, undermining its intended health benefits.

Furthermore, a striking finding was the infrastructure deficit relative to student enrollment. Observations revealed that the available WASH facilities fell far short of national standards. According to the Ghana Education Service's national school sanitation framework, the recommended ratio is one latrine pit per 50 students. In Mfantseman, however, ratios exceeded 1: 50 in some rural schools, forcing overcrowding and accelerated wear of existing facilities. This significant shortfall points to a systemic failure in aligning infrastructure provision with student population growth.

2.3 Operations and maintenance of WASH facilities in schools

Aladago et al. [4], Mohammed et al. [19] found that utilisation of school WASH facilities is associated with their level of cleanliness. Also, studies by [2] and [11] discovered that schools with well-maintained WASH facilities had higher student attendance, academic

Table 4 Distribution of toilet, water, and handwashing facilities across 16 basic schools in Mfantseman Municipality

School name	Toilets	Water	Handwashing
Eguaase Anglican Basic School	Yes	No	Yes
Baifikrom Methodist 'A' Basic School	Yes	Yes	Yes
Yamoransa M/A Basic School	Yes	No	Yes
Akatekyiwa St. Micheal Basic School	Yes	Yes	Yes
Asafara Catholic Basic School	Yes	No	Yes
Biriwa Methodist 'A' Basic School	No	Yes	Yes
Anomabo Anglican 'B' Basic School	Yes	No	Yes
Taido M/A Basic School	Yes	No	Yes
Egyaa M/A Basic School	Yes	Yes	Yes
Abandze Methodist 'A' Basic School	Yes	Yes	Yes
Kormantse M/A Basic School	Yes	Yes	Yes
I.T. Ahmadiyya Basic School (Saltpond)	No	Yes	Yes
Abonko M/A Basic School	Yes	Yes	Yes
Hinii M/A Basic School	No	No	Yes
Kuntu M/A Basic School	Yes	Yes	Yes
Edumadzie 'B' Basic School	Yes	No	Yes

Source: Field survey (2021)

achievement, and lower incidence of diarrheal illness. Therefore, the study collected and presented information on the operations and maintenance of WASH facilities in basic schools within the Mfantseman Municipality.

2.4 People responsible for cleaning the WASH facilities

The results showed that the operation and maintenance of WASH facilities were primarily handled by students, supervised by teachers. Operations and maintenance (O&M) are the day-to-day management of the facilities, which contribute to the long-term management goal. In both urban and rural schools, over 98% of cleaning duties were carried out by pupils, with negligible involvement of paid staff. Cleaning was organized through duty rosters, though in some schools it was also used as a form of punishment. This reliance on pupil labour mirrors findings by [19], who linked inadequate WASH support to increased absenteeism.

The findings of this study revealed that the operation and maintenance (O&M) of WASH facilities in basic schools within the Mfantseman Municipality are primarily handled by pups, under the supervision of teachers. In both rural and urban schools, over 98% of the cleaning activities were carried out by students, with minimal involvement of paid janitorial staff. Cleaning responsibilities were often organized through duty rosters, although in some cases, they were also used as a form of disciplinary action. This heavy reliance on pupil labour reflects a broader pattern noted in similar Ghanaian studies [4, 19], where limited institutional funding and resource allocation force schools to adopt self-help strategies to sustain hygiene operations.

In line with Ghana’s WASH policy, teachers are expected to supervise cleaning routines as part of school health education programs, while students participate in cleaning activities to promote practical hygiene learning and responsible behaviour [14, 23]. However, the study found that this practice has sometimes been misapplied, where cleaning is used as a form of punishment rather than a structured hygiene education exercise. This undermines the policy’s intention of using participatory cleaning as an avenue for environmental stewardship and behavioural change (Table 5).

The findings are supported by the in-depth interviews as follows:

The facilities are cleaned by the students while the teachers supervise [49-year-old head teacher from urban school]
Students do the cleaning while teachers supervise, but G.E.S. provides the detergents, soaps, and tissue papers to the schools [A 46-year-old head teacher from a rural school].

2.5 How cleaners are recruited to clean the WASH facilities

The findings from the study revealed that students are mainly appointed to clean the WASH facilities based on a dusty roster and sometimes as punishment to offenders in both rural and urban schools, respectively. The findings are illustrated in Table 6.

Table 5 People responsible for cleaning the WASH facilities

Variables	Students	Pain worker	Total
Rural	175 (100%)	0 (0)	175(100%)
Urban	158(98.1%)	3 (1.9%)	161 (100%)

Source: Field survey (2021)

Table 6 How cleaners are recruited to clean the WASH facilities

Variables	Punishment	Paid worker	Duty roster	Total
Rural	38 (21.7%)	0 (0)	137 (78.3%)	175 (100%)
Urban	38 (23.6%)	1 (0.6%)	122 (75.8%)	161 (100%)

Source: Field survey (2021)

Table 7 People responsible for providing detergent to clean the WASH facilities

Variables	Rural		Urban		Total		ASYMP.SIG
	Freq	%	Freq	%	Freq	%	
PTA	4	2.2	8	4.3	12	3.3	0.006
GES	148	80.4	134	72.8	282	76.6	
P.individual	9	4.9	10	5.4	19	5.2	
Don't know	4	2.2	20	10.9	24	6.5	
Sch. Authority	19	10.3	12	6.5	31	8.4	
Total	184	100	184	100	368	100	

Source: Field data (2021)

The findings are supported by the in-depth interview conducted:

It is a general norm in this school that the last Section in the week will clean the facilities, hence the students must do so if their Section is last. [A 46-year-old head teacher from a rural school]

Students clean the facilities and it is required as their duty and sometimes too as punishment [43-year-old head teacher from rural school]

School health team which is made up of teachers and prefect supervise, while students do the cleaning based on duty roster [53-year-old head teacher from urban school]

2.6 People responsible for providing detergents to clean the WASH facilities

Table 7 presents information on responsible agencies or individuals who provide logistics to schools for operation and maintenance of the WASH facilities. It was discovered that the Ghana Education Service (GES) is the main provider of detergents to schools in both the rural (80.4%) and urban (72.8%) areas for cleaning purposes, and sometimes through the support from Parents and Teachers Association (PTA) now (Parents Association), philanthropist and internally generated funds through school worships or charcoal selling/firewood business.

In Ghana, school sanitation and water management practices are guided by several policy frameworks, including the Ghana's National Sanitation Policy 2015 as referenced by [16], the [13] and [14]. These frameworks emphasize shared responsibility among key stakeholders, the Ghana Education Service (GES), school authorities, Parent Associations (PAs), and local communities for ensuring functionality, maintenance, and behavioural hygiene education in schools. Under these policies, the GES is responsible for coordinating resource allocation through the Capitation Grant, which is intended to support the overall operation of public schools, including the maintenance of WASH facilities. However, this grant is often insufficient, as confirmed by headteachers during interviews, forcing schools to rely on other funding from PAs, philanthropists, and income-generating activities such as the sale of firewood or charcoal to supplement maintenance costs.

Regarding infrastructure management, the construction of WASH facilities in public schools falls under the mandate of the Ghana Education Service, often in collaboration with the District Assemblies and the Ministry of Sanitation and Water Resources, while private schools are responsible for their own facility development. The procurement of spare parts and minor repairs, however, is the direct responsibility of individual schools, financed through the Capitation Grant or contributions from Parent Associations. Major repairs and rehabilitation are typically expected to be supported by the District Assemblies or through special intervention programmes such as those under World Health Organization and the World Bank's sanitation projects.

The evidence from Mfantseman illustrates the operational gap between policy intent and school-level practice. While the national frameworks advocate for multi-stakeholder collaboration and daily operations and maintenance routines, in practice, financial constraints and limited technical guidance have transferred much of the maintenance burden to schools themselves, particularly to teachers and pupils. Teachers have adapted by supervising cleaning as a cost-saving strategy and as a way to inculcate hygiene values among students, but this approach remains inconsistent across schools. Therefore, rather than treating cleaning as punishment, it should be institutionalized as part of daily hygiene education and life-skills development, reinforcing both health and civic responsibility among pupils.

The findings are supported by the in-depth interview conducted with some head teachers:

Mostly the detergents and tissues papers are provided by G.E.S through the capitation grant and at times some philanthropist [56 years head teacher from rural school]

Because of Covid-19, the government through G.E.S have been providing tissue paper and detergents often and mostly through the capitation grant as well [49 years head teacher from urban school]

2.7 The frequency at which the facilities are cleaned or maintained

The findings from the study revealed that although schools reported that facilities were cleaned daily, detergents were typically used only three times per week, largely due to resource shortages. Specifically, 71.2% and 72.3% of pupils in both rural and urban schools, respectively, were of the view that the school WASH facilities are cleaned only three times per week. Further probing revealed that mostly, the Ghana Education Service was the main provider of detergents and cleaning supplies, supplemented occasionally by Parent Associations (PAs), philanthropists, or income-generating school activities. This echoes [7], who found that schools often lack adequate logistics for sustainable management of sanitation facilities (Table 8).

The findings are supported by the in-depth interviews as follows:

The facilities are cleaned with detergents three times a week, and it is a strategy for managing the detergents since we do not know the exact timelines to obtain disinfectants from GES [46-year-old head teacher from rural school]

The students sweep the facilities every day, but they scrub with detergents and water three times a week [48-year-old head teacher from urban school]

Table 8 Frequency at which the facilities are cleaned or maintained

Variables	Rural		Urban		Total		ASYMP.SIG
	Freq	%	Freq	%	Freq	%	
Every day	13	7.1	19	10.3	32	8.7	0.420
Three days	131	71.2	133	72.3	264	71.7	
Rarely	17	9.2	17	9.2	34	9.2	
Sometimes	23	12.5	15	8.2	38	10.3	
Total	184	100	184	100	368	100	

Source: Field data (2021)

Table 9 Means of handling sanitary waste at school during menstruation

Variables	Rural		Urban		Total		ASYMP.SIG
	Freq	%	Freq	%	Freq	%	
Separate bin	3	1.6	2	1.1	5	1.4	0.459
No bin	165	89.7	173	94	338	91.8	
Use usual bin	7	3.8	3	1.6	10	2.7	
Don't know	9	4.9	6	3.3	15	4.1	
Total	184	100	184	100	368	100	

Source: Field data (2021)

2.8 Means of handling sanitary waste at school during menstruation

Menstrual hygiene management was identified as a critical gap. Most schools lacked dedicated bins for sanitary waste disposal, leaving female students without adequate support during menstruation. Thus, about 165 (89.7%) and 173 (94%) pupils, respectively in both rural and urban schools, were of the view that there are no separate bins in schools purposely for handling menstrual hygiene waste. [5, 19, 24] all confirm that inadequate menstrual hygiene facilities directly contribute to poor attendance and gender inequities in education (Table 9).

2.9 Availability of menstrual hygiene facilities in basic schools

Among the sixteen schools surveyed, only Baifikrom Methodist 'A' Basic School had a separate facility dedicated to menstrual hygiene management for female students. The absence of such facilities in the remaining schools represents a serious concern. Menstrual hygiene is not only a health requirement but also a critical factor influencing the dignity, confidence, and academic engagement of female students. Without access to safe and private spaces to manage menstruation, many girls are likely to experience absenteeism, discomfort, and reduced concentration in class. Research in Ghana and other developing contexts has shown that inadequate menstrual hygiene facilities contribute to higher dropout rates among female students and negatively affect their participation in learning activities [19].

The lack of menstrual hygiene infrastructure therefore threatens both the health and educational outcomes of female pupils. Girls without appropriate facilities may resort to unsafe alternatives that increase their risk of infections, while also suffering from stigma and embarrassment during menstruation. This creates an environment of inequality where female students are disadvantaged compared to their male peers. For the Mfantseman Municipality, addressing this gap is urgent, as it directly impacts progress toward gender equity in education and the broader Sustainable Development Goals (SDG 3 on good health and well-being, and SDG 4 on quality education) (Table 10).

Table 10 Availability of menstrual hygiene facilities in basic schools

School Name	Available menstrual hygiene facility
Eguaase Anglican Basic School	No
Baifkrom Methodist 'A' Basic School	Yes
Yamoransa M/A Basic School	No
Akatekyiwa St. Micheal Basic School	No
Asafara Catholic Basic School	No
Biriwa Methodis 'A' Basic School	No
Anomabo Anglican 'B' Basic School	No
Taido M/A Basic School	No
Egyaa M/A Basic School	No
Abandze Methodist 'A' Basic School	No
Kormantse M/A Bsic School	No
I.T. Ahmadiyya Basic School (Saltpond)	No
Abonko M/A Basic School	No
Hinii M/A Basic School	No
Kuntu M/A Basic School	No
Edumadzie 'B' Basic School	No

Source: Field survey (2021)

Table 11 Level of satisfaction in maintenance and operation of WASH facilities

Variables	Rural		Urban		Total		ASYMP.SIG
	Freq	%	Freq	%	Freq	%	
Very satisfied	5	2.7	12	6.5	17	4.6	0.180
Satisfied	56	30.4	68	37.0	124	33.7	
Neutral	40	21.7	38	20.7	78	21.2	
Dissatisfied	68	37.0	52	28.3	48	13.0	
Very dissatisfied	15	8.2	14	7.6	29	7.9	
Total	184	100	184	100	368	100	

Source: Field data (2021)

2.10 Level of satisfaction in maintenance and operation of WASH facilities

The findings revealed that there were notable differences between urban and rural schools in student satisfaction. About 37% of urban pupils expressed satisfaction with facility management, compared with an equal proportion of rural pupils who reported dissatisfaction. Observations confirmed that rural schools often had broken doors, poor roofing, and inadequate water supply (Figs 8 and 9). The views of the respondents are presented in Table 11. Nkonde et al. [21] argue that such systemic deficits in school sanitation highlight structural inequalities across Ghana's education system.

The findings are supported by responses from some stakeholder heads as follows:

I am not satisfied at all with how the facilities in the schools are maintained because government capitation is not adequate; however, it is not for facility maintenance alone. Hence, inadequate funds sometimes affect our ability to ensure the maintenance of the facilities. [A 36-year-old stakeholder]

I am not satisfied because even though school health education programs are organised frequently with the help of some health personnel on their worship days, some students act irresponsibly sometimes and mess up the facilities. [48-year-old stakeholder]

No matter how well you clean the place, you will come back to school the following day and realize that the community members have broken the locks and messed up the place. [56-year-old stakeholder]

2.11 Evidence of state and condition of WASH facilities in schools

Figure 8 shows the state of WASH facilities used in the basic schools. It typically shows urinal with broken doors, implying a lack of privacy among students when they want to urinate.

The picture below shows the state of WASH facilities used in the basic schools. It typically shows a toilet facility with a bad roof, implying an uncomfortable feeling among students when they want to ease themselves.

Objective 2: Explore the barriers to effective WASH facilities management.

2.12 Challenges facing WASH facilities management

Challenges reported included inadequate funding, poor institutional support, irresponsible student behaviour, and community misuse of school facilities. Several schools reported that community members vandalized facilities, reflecting wider sanitation deficits in local households. Nunbogu et al. [22] highlight how weak WASH systems are intertwined with broader socio-cultural and gendered dynamics in Ghana (Figs. 7, 8 and 9). The p -value of 0.024 indicates that there is a statistically significant difference between rural and urban areas in terms of challenges faced in managing WASH facilities in basic schools. This implies that, challenges exist in the basic schools, but the magnitude of the challenge differs from one school to another based on different factors. The information is presented in Table 12 below.

The picture below shows the state of WASH facilities used in the basic schools. It typically shows condition of toilet facilities, implying an uncomfortable feeling among students when they want to ease themselves.



Fig. 7 State of WASH facilities (urinal with broken doors).

Source: Field Survey (2021)



Fig. 8 State of WASH facilities (toilet with bad roofing).
Source: Field Survey (2021)



Fig. 9 State of WASH facilities (condition of the toilet in schools with unhygienic floors).
Source: Field Survey (2021)

2.13 Specific challenges in managing WASH facilities

The study solicited the views of respondents and participants to enquire whether some of the specified challenges are prevalent in the public basic schools in Mfantseman Municipality (Table 13). The findings from the study revealed that the majority

Table 12 Challenges facing WASH facilities management

Responses	Rural		Urban		Total		ASYMP.SIG
	Freq	%	Freq	%	Freq	%	
Yes	169	91.8	155	84.2	324	88	0.024
No	15	8.2	29	15.8	44	12	
Total	184	100	184	100	368	100	

Source: Field data (2021)

Table 13 Specific challenges in managing WASH facilities

Variable	Rural		Urban		Total		ASYMP.SIG
	Freq	%	Freq	%	Freq	%	
Lack of support from local institution							
SA	51	30.2	71	45.8	122	37.6	0.000
A	94	55.6	80	51.7	174	53.7	
D	19	11.2	3	1.9	22	6.8	
SD	5	3	1	0.6	6	1.9	
Lack of proper supervision by the school authority							
SA	0	0	7	4.5	7	2.2	0.001
A	12	7.1	24	15.5	36	11.1	
D	117	69.2	83	53.5	200	61.7	
SD	40	23.7	41	26.5	81	25	
Poor work habits							
SA	2	1.2	7	4.5	9	2.8	0.000
A	26	15.4	52	33.5	78	24.1	
D	107	63.3	70	45.2	177	54.6	
SD	34	20.1	26	16.8	60	18.5	
PTA's unwillingness to pay							
SA	42	24.9	35	22.6	77	23.8	0.911
A	110	65	101	65.2	211	65.1	
D	16	9.5	18	11.6	34	10.5	
SD	1	0.6	1	0.6	2	0.6	
Irresponsible student behaviour							
SA	10	6	10	6.5	20	6.2	0.320
A	72	42.6	77	49.7	149	46	
D	68	40.2	47	30.3	115	35.5	
SD	19	11.2	21	13.5	40	12.3	
Community invasion							
SA	91	53.8	57	36.8	148	45.7	0.000
A	61	36.1	75	48.4	136	42	
D	17	10.1	11	7.1	28	16.6	
SD	0	0	12	7.7	12	3.7	
Odour							
SA	42	24.8	57	36.8	99	30.6	0.000
A	93	55	90	58	183	56.5	
D	29	17.2	8	5.2	37	11.4	
SD	5	3	0	0	5	1.5	
Inadequate water							
SA	65	38.5	40	25.8	105	32.4	0.000
A	56	33.1	79	51	135	41.7	
D	34	20.1	31	20	65	2	
SD	14	8.3	5	3.2	19	5.9	
Total	169	100	155	100	324	100	

Source: Field data (2021) [SA = Strongly agree, A = Agree, D = Disagree, SD = Strongly disagree]

of students in both the rural and urban areas either strongly agreed or agreed with the fact that there is a lack of support from local institutions to manage the school facilities. The p-value of 0.000 indicates that there is a statistically significant difference between rural and urban areas when it comes to getting support from local institutions to manage the school facilities. It implies that the magnitude of challenges faced and the support to overcome them differ from one school to another. The findings are supported by responses as follows:

Most schools in the Municipality are really facing challenges in operating and maintenance of the facilities, but the problem is inadequate funds, and this usually occurs because the common fund for initiating development doesn't come regularly [48-year-old Stakeholder head].

At first government, through the District Assembly, was paying the water bills, but they are no longer paying; as such, the taps are disconnected due to huge debt. Hence, maintenance of the WASH facilities is a major problem, making the facility unhygienic to use [48-year-old stakeholder]

Objective 3: Propose evidence-based and policy-relevant strategies for improving facility maintenance and ensuring reliable access to safe water, sanitation, and hygiene services.

2.14 Strategies to improve utilisation and maintenance of WASH facilities

Findings from the study revealed that the majority of students in both the rural and urban areas either strongly agreed or agreed with the statement that promoting hygiene education, institutional support, policy enforcement, and improving WASH quality and student-latrine ratio are some of the strategies that can help to improve utilisation and management of the school WASH facilities (Table 14).

The findings are in line with the in-depth interview with the stakeholder heads:

I believe repairing damaged facilities and making it compulsory to use without any other means of waste disposal at schools will help control unsanitary behaviour among pupils [48-year-old stakeholder head].

I think if government in collaboration with G. E. S., makes adequate funds available, it will help increase WASH facilities access and improve facility maintenance [51-year-old Head teacher from rural school].

3 Discussion

This study deepens understanding of Water, Sanitation, and Hygiene (WASH) management in public basic schools within Ghana by revealing that reported facility coverage in the Mfantseman Municipality conceals serious gaps in service quality, reliability, and inclusivity. Although most schools indicated the presence of toilets and handwashing stations, the absence of reliable on-premises water and inadequate maintenance rendered these facilities partially functional or symbolic rather than effective. These findings corroborate global and regional evidence that many schools fail to meet the minimum Joint Monitoring Programme (JMP) standards for WASH facilities, thereby compromising health and learning outcomes [18, 27–29].

Table 14 Strategies to improve utilisation and management of WASH facilities

Table 14 Strategies to improve utilisation and management of WASH facilities							
Variables	Rural		Urban		Total		ASYMP.SIG
	Freq	%	Freq	%	Freq	%	
Institutional support							
SA	91	49.5	90	48.9	181	49.2	0.598
A	92	50	94	51.1	186	50.5	
SD	1	0.5	0	0	1	0.3	
Policy enforcement							
SA	89	48.4	70	38	159	43	0.046
A	95	51.6	114	62	209	57	
Regular maintenance of facilities							
SA	75	40.8	87	47.3	162	44.1	0.185
A	107	58.2	97	52.7	204	55.4	
D	2	1	0	0	2	0.5	
Hygiene education							
SA	82	44.6	97	53	179	48.6	0.911
A	102	55.4	86	46.7	188	51.1	
D	0	0	1	0.5	1	0.3	
Improve WASH quality and student-latrine ratio							
SA	85	46.2	91	49.5	176	47.8	0.054
A	99	53.8	88	47.8	187	50.8	
D	0	0	5	2.7	5	1.4	
Total	184	100	184	100	368	100	

Source: Field data (2021) [SA=Strongly agree, A=Agree, D=Disagree, SD=Strongly disagree]

The predominance of Kumasi Ventilated Improved Pit (KVIP) latrines and the high share of partially functional units point to persistent infrastructure and service-quality gaps, a trend already highlighted by [1, 4, 6]. Schools without toilet facilities were observed to depend on beaches or bushes for defecation, practices that expose pupils to infections, undermine dignity, and increase absenteeism, particularly among girls. Comparable findings were reported by [20] in rural Tanzania, where the absence of private and hygienic facilities significantly affected girls' school attendance and menstrual hygiene management. These realities mirror [26] evidence that districts with poor sanitation coverage also record lower educational outcomes.

Water availability proved the principal constraint to sustainable sanitation and hygiene. In several schools, water was fetched from nearby houses, purchased from vendors, or carried from home, conditions inconsistent with the definition of basic service levels under JMP standards [27, 28]. Similar patterns of limited access to safe and reliable water have been observed in Sierra Leone and East Africa, where inadequate infrastructure and maintenance undermine community hygiene and school health outcomes [18, 25]. The irregular supply of water and cleaning materials in Mfantseman therefore undermines daily sanitation routines and contributes to unhygienic conditions comparable to those documented by [11, 31].

Operations and maintenance (O&M) practices also reflect systemic weaknesses. The study found that cleaning was largely undertaken by pupils under teacher supervision, with detergents supplied irregularly by the Ghana Education Service (GES) and supplemented by Parent Associations and philanthropists. Although this arrangement sustains minimal functionality, it shifts institutional responsibility to pupils and also, using cleaning as punishment rather than as part of structured hygiene education undermine the primary goal of inculcating hygiene education and civic responsibility into students'

behaviour. This tendency reinforces findings from [11, 24] that weak supervision and poor resource allocation erode school hygiene culture. The infrequent use of detergents, about three times weekly due to limited resources, further demonstrates how constrained operations and maintenance budgets directly affect facility cleanliness and usability [7].

Gender inequities emerged as particularly severe. Only one of the sixteen schools surveyed possessed a dedicated menstrual hygiene management (MHM) facility, and over ninety percent of pupils reported the absence of sanitary bins. This deficiency curtails girls' comfort, safety, and participation, reinforcing evidence that poor MHM infrastructure contributes to absenteeism, stigma, and diminished academic performance [1, 2, 5, 7, 19, 20, 24].

Finally, the study exposes contradictions in community engagement. While collaboration is often promoted as key to sustainability, several schools reported vandalism and misuse of facilities by community members, confirming that local sanitation deficits and weak institutional coordination undermine school WASH systems [21, 22]. Inadequate supervision, inadequate institutional support (Municipal Assembly and Ghana Education Service), and poor behaviour further compound these challenges. Consequently, the latrine-to-student ratios far exceed the national benchmark of 1:50, signaling deep structural misalignments between enrolment growth, infrastructure planning, and enforcement of national sanitation standards.

3.1 Implications of findings

The findings have important implications for both schools and the municipality. At the school level, irregular water supply and substandard sanitation expose pupils to preventable infections and reduce effective learning time, as students spend hours coping with water shortages or avoiding unhygienic facilities. These outcomes not only depress academic performance but also raise long-term health risks. For girls, the lack of menstrual hygiene infrastructure and safe disposal systems fosters absenteeism, social exclusion, and declining confidence, barriers to achieving gender equity in education.

At the municipal and institutional level, the results underscore the need for predictable funding and robust governance mechanisms. Without dedicated operations and maintenance (O&M) budgets, even well-constructed facilities deteriorate quickly, diminishing returns on infrastructure investment. The findings also suggest that the sustainability of WASH services depends on integrating community engagement with protection measures such as fencing, lockable facilities, and the provision of alternative public toilets to curb vandalism. Ultimately, alignment with national policy frameworks, the Ghana's National Sanitation Policy 2015 as referenced by [16], and [13], is critical to standardize practices and ensure accountability.

3.2 Contribution to existing knowledge

This study enriches existing WASH literature by providing empirical evidence on how water, sanitation, and hygiene (WASH) facilities are managed across public basic schools in the Mfantseman Municipality, a context that has received little scholarly attention despite its critical sanitation and educational challenges.

It extends prior research [3, 4, 11] by combining quantitative, qualitative, and visual evidence, thereby offering a more comprehensive understanding of how infrastructural

availability, institutional support, and behavioural factors interact to shape WASH management outcomes.

The study contributes theoretically by applying and validating the Sanitation Behaviour Change Framework (SaniFOAM) in a school setting, moving beyond household-level sanitation research to institutional contexts. This demonstrates how opportunity, ability, and motivation collectively influence sanitation behaviour and facility maintenance within learning environments.

It contributes to gender-focused WASH scholarship [5, 19] by exposing the neglect of menstrual hygiene management (MHM) in basic schools and showing how such deficiencies perpetuate absenteeism, stigma, and gender inequality in education.

The study provides policy-relevant insights by revealing implementation gaps between Ghana's sanitation policies, such as the Ghana's National Sanitation Policy 2015 as referenced by [16], and [14], and actual school-level practices. This bridges the divide between policy frameworks and ground-level realities, offering evidence-based recommendations for improvement.

Finally, the integration of photographic and observational evidence represents a methodological contribution, demonstrating how visual ethnography can complement mixed-method research to document and communicate the lived realities of sanitation management in educational institutions.

4 Conclusions

This study concludes that WASH management in public basic schools within the Mfantseman Municipality is characterized by uneven infrastructure, weak maintenance systems, and persistent gender inequities. Although basic toilets and handwashing stations exist in most schools, their effectiveness is constrained by unreliable water supply, overcrowded usage, and inadequate cleaning logistics. The overreliance on pupil labour for operations and maintenance reflects systemic under-institutionalization, while the near absence of menstrual hygiene facilities amplifies inequalities in participation and learning. Theoretically, the study demonstrates the value of the SaniFOAM framework in revealing how opportunity, ability, and motivation interact to shape WASH behaviours in school settings. Practically, the evidence underscores an urgent need for sustained investment, behavioural reinforcement, and policy enforcement to translate infrastructure into meaningful service delivery.

4.1 Recommendations

To address the identified challenges, the study recommends a multi-level response that combines infrastructure upgrading, institutional strengthening, and behavioural interventions.

First, the Ghana Education Service and municipal authorities should ensure that all schools meet the national standard of one latrine per fifty students, prioritizing schools with the most severe shortfalls. Adequate and reliable water supply must be guaranteed through piped connections, boreholes, or protected standpipes equipped with storage and routine functionality checks, in line with [28] service classifications.

Second, gender-sensitive infrastructure must be mainstreamed into school construction and rehabilitation projects. Every basic school should include private, safe, and

well-equipped menstrual hygiene management spaces with water, soap, and disposal bins, accompanied by regular hygiene education and teacher training.

Third, operations and maintenance financing should be institutionalized by ring-fencing funds within the capitation grant for cleaning materials, minor repairs, and sanitary waste management, while introducing simple facility registers to track cleaning frequency, repair schedules, and functionality status.

Cleaning duties should be repositioned from punitive activities to structured hygiene practice, coordinated by trained teachers and student health clubs.

Finally, community-school collaboration should be strengthened through local bylaws and education campaigns that emphasize shared responsibility while discouraging misuse of facilities. Installing fences and locks, constructing alternative community toilets, and involving local leaders in monitoring will help protect school assets.

At the policy level, there should be incorporating of [28] service ladders into national Education Management Information System (EMIS) reporting and disaggregating data by gender and rural–urban location will enhance targeting of support and equity monitoring.

4.2 Limitations of the study

While this study provides valuable insights into the management of Water, Sanitation, and Hygiene (WASH) facilities in public basic schools within the Mfantseman Municipality, several limitations should be acknowledged.

First, the research was geographically limited to sixteen public basic schools in one municipality. Although these schools were systematically selected to represent both rural and urban settings, the findings may not be fully generalizable to all schools in Ghana, particularly private institutions or those in different ecological or socio-economic contexts.

Second, the study relied primarily on self-reported data from pupils, headteachers, and institutional stakeholders. Despite efforts to ensure honesty and confidentiality, self-reporting can introduce bias, as some participants may have provided socially desirable responses, especially on sensitive issues such as open defecation, menstrual hygiene, and maintenance practices.

Third, while structured observation complemented the quantitative data, the study's cross-sectional design captured only a snapshot of facility conditions at a single point in time (February 2021). As a result, seasonal variations in water supply or changes in maintenance practices over time were not assessed. A longitudinal or follow-up study could provide a more dynamic understanding of WASH management trends.

Fourth, the study did not include detailed cost analysis of operation and maintenance (O&M) activities or explore the economic efficiency of various funding mechanisms, such as the Capitation Grant or community contributions. Understanding these financial dimensions would be essential for designing sustainable policy interventions.

Finally, the study's photographic documentation, though anonymized and ethically approved, may not capture all aspects of facility functionality or user experience. Observations were limited to visible infrastructure conditions and did not extend to microbial water quality testing or detailed engineering assessments.

Despite these limitations, the study offers a robust and contextually grounded understanding of how WASH facilities are managed in basic schools within the Mfantseman

Municipality. The methodological rigor, combination of mixed methods, and triangulation of data sources provide confidence in the reliability of the findings and their value for guiding future research, policy, and practice.

4.3 Suggestions for future research

Future research should include cost–benefit analyses to evaluate the sustainability of existing funding mechanisms, such as the Capitation Grant and community contributions to WASH facilities maintenance.

Also, future studies should explore the behavioural and socio-cultural factors influencing WASH practices among pupils and teachers.

Finally, future research should incorporate water quality testing and hygiene audits to provide stronger evidence on health outcomes linked to school sanitation conditions.

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Authorship

The authors confirm that they have read and understood the journal's authorship and publication policies, and that this manuscript is being submitted in full accordance with those policies.

Disclaimer

The findings and conclusions presented in this report are based on data collected from a limited number of selected schools. As such, they may not necessarily represent the situation across all basic schools in the Mfantseman Municipality. The interpretations should therefore be understood within the context of the specific schools studied.

Author contributions

Annan, K., K.: Contributed to the conceptualisation, methodology, data collection and entry, data analysis and interpretation, writing and review of the paper. Mariwah, S.: Contributed to the conceptualisation, methodology, writing and review of the paper. Annan, A.: Contributed to the conceptualisation, methodology, writing, and review of the paper. Chirisa, I.: Contributed to the conceptualisation, methodology, writing and review of the paper.

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Data availability

The datasets supporting the findings of this study are not publicly available due to confidentiality agreements with participating schools and the Ghana Education Service. However, data may be made available upon reasonable request to the corresponding authors [**annankingsfordkobina@gmail.com](mailto:annankingsfordkobina@gmail.com); smariwah@ucc.edu.gh and with formal permission from the Ghana Education Service, Mfantseman Division (**ges@mma.edu.gh).

Declarations

Ethics approval and consent to participate

The research was approved by the Institutional Review Board (IRB) of the University of Cape Coast, under approval number UCCIRB/CHLS/2021/78, in accordance with the IRB's guidelines and the ethical principles outlined in the Declaration of Helsinki. The study also complied with Ghana's Data Protection Act, 2012 (Act 843), ensuring the confidentiality and responsible handling of participants' personal data. Permission to conduct the research in selected schools was formally granted by the Ghana Education Service (GES) office in the Mfantseman Municipality, following submission of an introductory letter. Written informed consent was obtained from all participants. For minors (those under 18 years), parental or guardian consent was additionally secured, in line with Article 28, Clause 5 of the 1992 Constitution of Ghana. Teachers were also briefed on the study's purpose and procedures. Participants were assured of anonymity, confidentiality, and the voluntary nature of participation, including their right to withdraw at any time without penalty.

Consent for publication

Consent for publication of photographs and images was obtained from all relevant participants. For pupils under the age of 18, written consent was additionally secured from parents or legal guardians, as well as permission from the Ghana Education Service in the Mfantseman Municipality. All images were taken strictly for academic purposes and were

not intended to facilitate the easy identification of participants. Every effort was made to anonymize the photographs and ensure that individuals cannot be personally recognized.

Competing interests

The authors declare no competing interests.

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